

Name of the experiment: Simplify and implement the following logic expression:

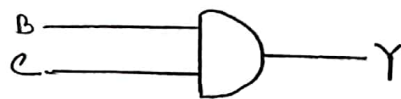
$$Y = \bar{A}BC + ABC + ABC$$

Objective: The objective of this experiment is to simplify and implement the given expression.

Theory:

$$\begin{aligned}\text{Let, } Y &= \bar{A}BC + ABC + ABC \\ &= \bar{A}BC + ABC \quad [A + A = A] \\ &= BC(\bar{A} + A) \\ &= BC \quad [\bar{A} + A = 1]\end{aligned}$$

Logic circuit



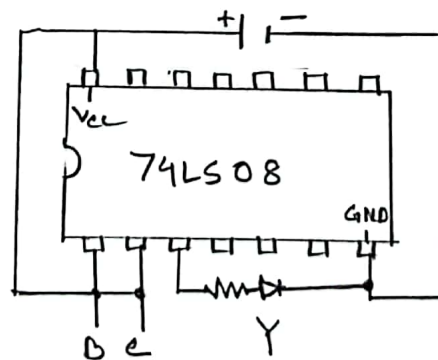
Truth table:

B	C	$Y = B \cdot C$
0	0	0
0	1	0
1	0	0
1	1	1

Apparatus:

- 1) Breadboard
- 2) Resistor, LED
- 3) wire
- 4) AND gate
- 5) Power supply.

Procedure: At first we have to identify the pin number of the IC and set it on breadboard. All the Vcc and GND should be connected properly, with the power supply. Output pin is connected to the resistor and then to the LED as shown below.



Result

B	C	BC	LED
0	0	0	off
0	1	0	off
1	0	0	off
1	1	1	on

Discussion

In this experiment we have checked all the pin of IC's and found out the active pins. We have also found out the position at which the IC's have power supply.

finally we have verified the experiment successfully.